Sprint Progress Report 3

Sprint Overview

During Sprint 3, our team focused on completing the user interface requirements through paper prototyping and organizing development tasks using GitLab issues. We also established a clear work division across frontend, backend, and database components to prepare for implementation.

1. Class Presentation

Our team completed the in-class presentation requirement

2. Work Completed

Frontend Development:

- Developed the functional frontend UI for the authenticated Dashboard, displaying total balances and recent activity.
- Built the "Add Expense" form interface, allowing users to input descriptions, amounts, and select participants.

Database Architecture & Server Setup:

- Provisioned the database server environment and successfully executed the SQL scripts to create the physical database tables on the server.
- Finalized the comprehensive MySQL schema design (including users, groups, group_members, expenses, and expense_splits tables). Implemented the ORM (Object-Relational Mapping) models in Python (/app/models) to seamlessly bridge our MySQL data with the backend code.

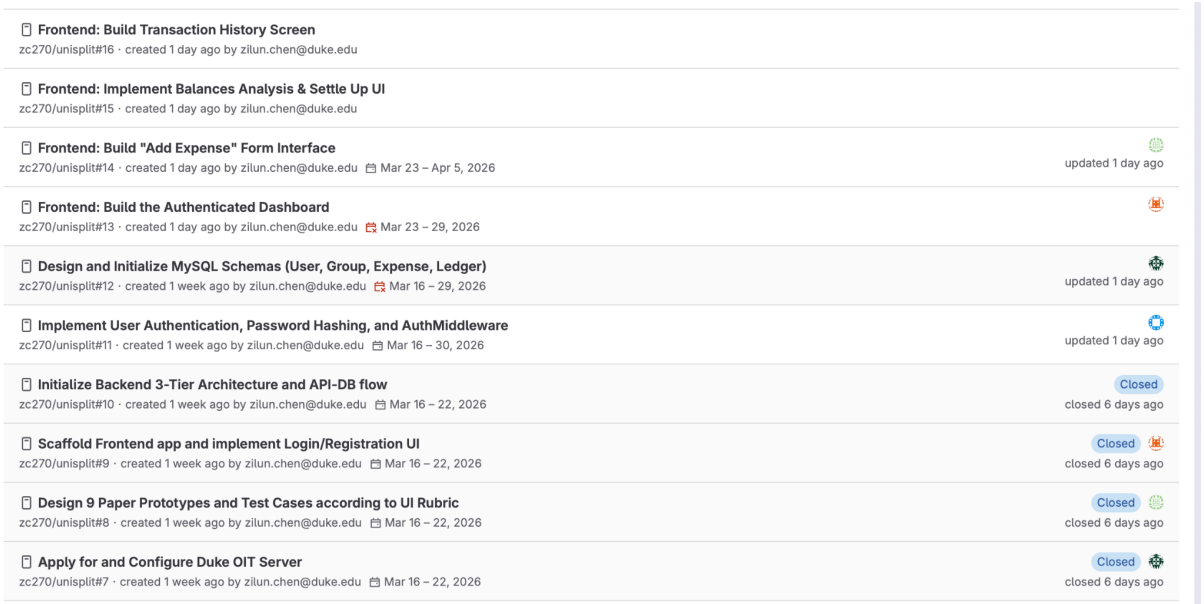
Core Backend Business Logic:

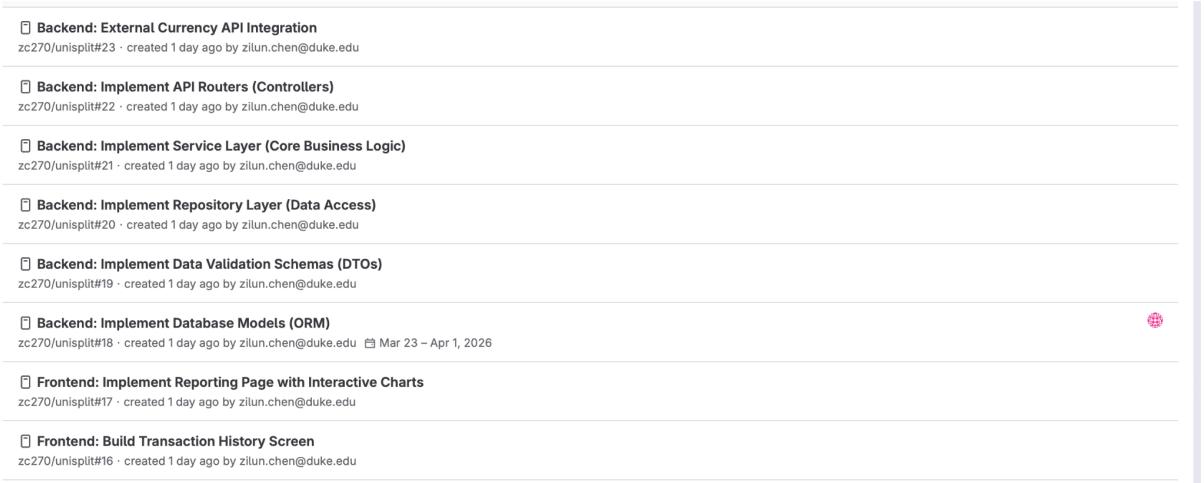
- Authentication Flow: Implemented secure user registration and login using bcrypt for password hashing and salting. Added validation to return 400 errors for duplicate emails.
- Group Management: Built the logic to assign group creators as administrators in the group_members table and created an invite system that validates existing users. Integrated an activity_log to track these actions.
- Expense Splitting Logic: Developed the mathematical routing to divide total expense amounts by the number of participants, dynamically writing individual debt shares into the expense_participants records.
- Balance Optimization Algorithm: Implemented a greedy algorithm that iterates through group expenses to calculate individual net balances, intelligently matching the largest debtors with the largest creditors to settle all debts using the absolute minimum number of transactions.

3. GitLab Issues & Work Division

We continued to use GitLab issues to track our backend and frontend tasks. Tasks were assigned as follows:

- **Frontend:** Build Dashboard, Build "Add Expense" Form Interface (Chenyao An, Kehan Zhu)
- **Database & ORM:** Design MySQL Schemas, Implement Database Models (ORM), DB Server Setup (Zilun Chen, Jiakai Lin)
- **Backend Logic:** Implement User Auth & Hashing, Service Layer Logic for Splitting & Greedy Algorithm (Jiehe Wu)
- **Backlog & Future Tasks:** In addition to the assigned tasks, we have proactively created and categorized additional issues in the backlog. This allows team members responsible for specific domains (Frontend, Backend, Database) to claim and self-assign upcoming tasks as they complete their current workload, ensuring a smooth agile workflow.





4. Project Artifacts Updates

Our project artifacts were updated to reflect the current progress of the sprint:

- Updated our system architecture documentation to accurately reflect the final MySQL table definitions (users, groups, expenses, activity_log, etc.).
- Created the backend directory structure, specifically establishing the /app/models for ORM integration.
- Architecture Diagrams: Created and added a C4 Model diagram to provide a clearer, multi-level visual representation of our system architecture.
- Codebase Updates:
    - Updated the database initialization SQL scripts in the database/ directory.
    - Committed the core frontend code for the completed Dashboard and Add Expense pages into the frontend/ directory.

5. Weekly Meeting Summary

Held weekly meeting at Mar 29th with TA. We reviewed the newly implemented greedy algorithm for debt settlement, discussed how the frontend forms will pass JSON data to backend logic, and verified the successful connection to database server.

6. Challenges & Learnings

- Challenge: Translating complex financial relationships (like multi-person uneven splits) into efficient backend code was difficult.
- Challenge: Ensuring the Python ORM correctly mapped to our relational MySQL schema required careful debugging, especially regarding foreign key relationships between groups, expenses, and users.
- Learning: By utilizing a greedy algorithm, we learned how to optimize the BalanceOptimizationService to reduce user transaction friction, fulfilling our "Split Bills, Not Friendships" core value proposition.

7. Next Steps (Sprint 4 Plan)

- Complete the development of all remaining frontend UI screens based on our paper prototypes.
- Comprehensively connect the frontend applications to the backend REST APIs. This API integration will not be limited to the Dashboard and Add Expense pages, but will encompass the full data flow across all functional interfaces.
- Implement data validation schemas (DTOs) to secure the API against bad frontend inputs.
- Begin integrating the external Currency API to handle international student trip expenses.